University of Florida, Department of Mathematics

Combinatorics PhD Exam, August 2018

1. Find a generating function for the number of positive integer solutions to $x_1+2x_2+3x_3+4x_4+5x_5=n$.

2. Tickets numbered $\{1,2,\ldots,n\}$ are drawn from an urn at random one after the other without replacement. What is the probability that the number r is drawn on the k^{th} drawing?

Among a population of $n+1$ people, a rumor is spread at random. One person tells the rumor to a second, who in turn repeats it to a third person, etc. What is the probability that the rumor will be told k times without being repeated to any person?

3. Prove that there exists no simple, bipartite, planar graph with minimum degree at least 4.

4. Let μ be the largest eigenvalue of the adjacency matrix of a graph G and Δ the maximum degree of G. Prove that $\mu \leq \Delta$.

(Hint: Consider the largest, in absolute value, coordinate of a corresponding eigenvector.)

5. Show that a k-regular graph of girth 4 has at least $2k$ vertices, and a k-regular graph of girth 5 has at least k^2+1 vertices. Draw a 3-regular graph of girth 4 with $2\cdot 3=6$ vertices and a 3-regular graph of girth 5 with $3^2+1=10$ vertices.

6. Let f_n be the number of all compositions of the integer n into odd parts. Find the exponential order of the numbers f_n.

7. Recall that a permutation is called even if it has an even number of even cycles, and it is called odd if it has an odd number of even cycles. Also recall that a derangement is a permutation with no 1-cycles.

Let A_n be the number of all even derangements of length n, and let B_n be the number of all odd derangements of length n. Find an exact formula for $A_n - B_n$.

8. This problem has two parts.

(a) Prove that if a (v,k,λ) BIBD exists with block set $\mathcal{B}$ on a point set V, then the set $\mathcal{B}'=\{V\setminus B \mid B\in\mathcal{B}\}$ is a BIBD.

(b) Construct a $(7,4,2)$ BIBD.